ZDC autoresearch, Phase 6: the crystal-seeded ZDC photon finder

Find the ZDC photons the hadronic section loses, with the LYSO crystals in front of it

Started and closed 2026-11-09 · one slide per try · rebuilt from the run's result files

Phase 6 outcome — Y9 → Z8 on the MSF search and held-out halves

1486 → 1946
Λ found, search halves (+31 %)

1411 → 1865
held-out halves (z +18.0)

26.7 → 30.9 %
purity in the ±30 MeV window

0.124 → 0.114
−t σ68 at 9x275, GeV²

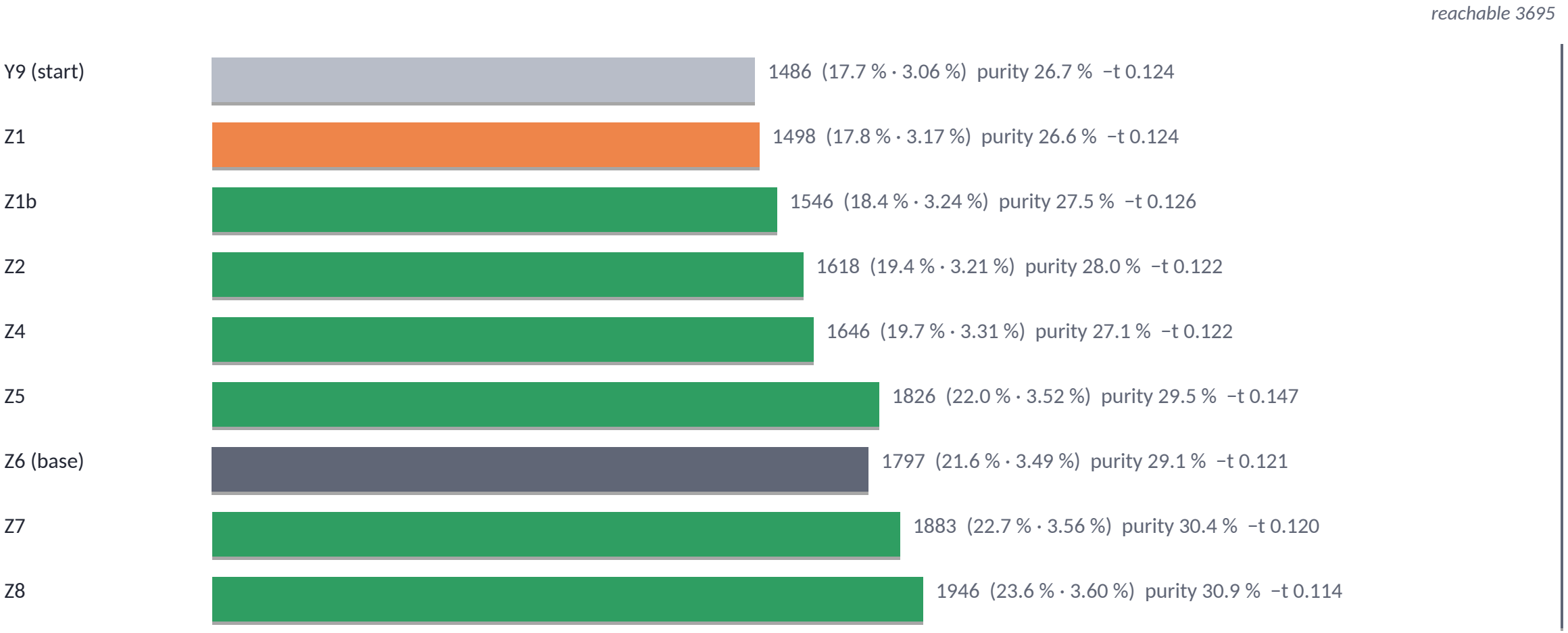
yield (found / expected)	Y9, start	Z8, final	reachable
9x275	17.70 %	23.58 %	43.6 %
9x130	3.06 %	3.60 %	8.0 %

What Z8 adds to Y9

- crystal-seeded photon candidates in the ZDC pairing (Z1b, Z2)
- a narrower Λ-mass term in the pairing (Z4)
- the crystal law's energy offered to Hcal photons (Z5), its column out of the neutron (Z6)
- an isolation term against neutron fragments (Z7)
- the axis-law energy for Hcal photons (Z8)

Z8 finds 23.6 % of expected at 9x275, 54 % of the reachable yield; closed 2026-11-09 (D-user-45).

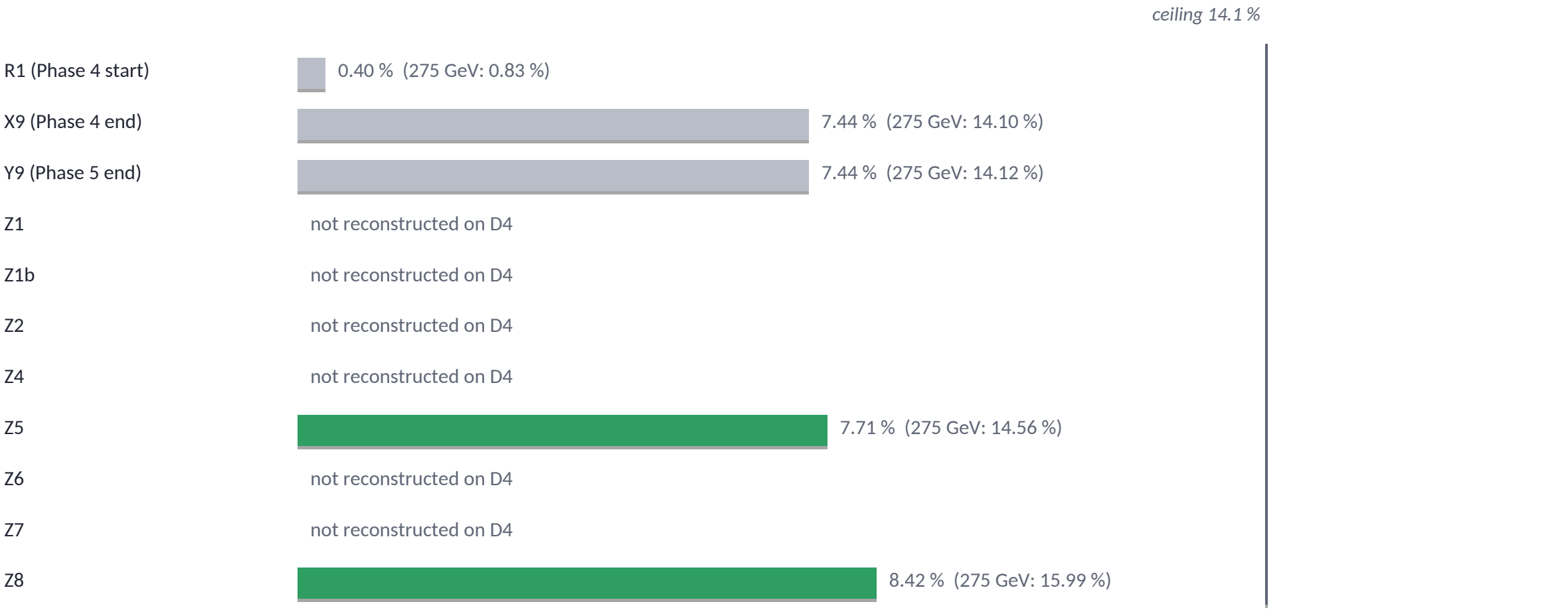
Score per try — Λ → nπ⁰ found on the MSF search halves, 9x275 + 9x130



Bar: found Λ pooled over the search halves (463). Text: found (yield 9x275 · 9x130), purity, −t σ68 at 9x275 in GeV² (476). Colour = verdict: green accepted, orange inconclusive, grey rejected. Z6 was rejected by the selector and made the base by the user for its −t (D-user-44). Rule: the reachable yield, pooled.

Y9 1486 → Z8 1946 found (+31 %): 53 % of the reachable yield.

D4 gun score per try — S4a_tm per thrown $\Lambda \to n\pi^0$, mean over 100–275 GeV



headline.csv (441). Phase 6 reconstructed the D4 gun at its two checkpoints only (Z5, Z8); D4 is reported, never selected on, and 410 credits Hcal clusters only (a crystal-seeded photon counts as missed).

On the gun, Z8 stands at 8.4 % of thrown (ceiling 14.1 %), above every Phase 4 and Phase 5 row.

Phase 6 tries — Λ found on the search halves, step by step

try	change	9x275	9x130	score	purity	z	−t σ68 9x275	verdict
Z1	crystal-seeded photons in the ZDC Λ pairing	1267 → 1271	219 → 227	1486 → 1498	26.7 → 26.6 %	+0.8	0.124 GeV²	INCONCLUSIVE
Z1b	crystal photons with the Hcal column from the rec hits	1267 → 1314	219 → 232	1486 → 1546	26.7 → 27.5 %	+3.6	0.126 GeV²	ACCEPTED
Z2	crystal photon parameters: threshold, column width, neutron share	1314 → 1388	232 → 230	1546 → 1618	27.5 → 28.0 %	+5.6	0.122 GeV²	ACCEPTED
Z4	the ZDC pairing's Λ-mass width 0.04 → 0.03 GeV	1388 → 1409	230 → 237	1618 → 1646	28.0 → 27.1 %	+3.2	0.122 GeV²	ACCEPTED
Z5	the crystal law's energy as a competing hypothesis for Hcal photons	1409 → 1574	237 → 252	1646 → 1826	27.1 → 29.5 %	+11.6	0.147 GeV²	ACCEPTED
Z6	the crystal-law photon's Hcal column leaves the neutron	1574 → 1547	252 → 250	1826 → 1797	29.5 → 29.1 %	-2.8	0.121 GeV²	REJECTED
Z7	tagged photons near the neutron core pay an isolation term	1547 → 1628	250 → 255	1797 → 1883	29.1 → 30.4 %	+7.7	0.120 GeV²	ACCEPTED
Z8	tagged Hcal photons may take the axis law's energy	1628 → 1688	255 → 258	1883 → 1946	30.4 → 30.9 %	+4.9	0.114 GeV²	ACCEPTED

Score 1486 → 1946 (+31 %) since the start row Y9; −t σ68 at 9x275 0.124 → 0.114 GeV² (Q22).

Phase 6 start row Y9 — yield, purity and −t resolution per beam

Y9 on the search halves D6s	9x275	9x130
N_found / N_expected	1267 / 7160	219 / 7160
yield (reachable)	17.70 % (43.6 %)	3.06 % (8.0 %)
purity in the ±30 MeV window	26.4 %	28.7 %
−t resolution σ68 of found Λ	0.12 GeV²	0.13 GeV²
−t within ±0.1 GeV²	56 %	58 %

Kept from Phase 5

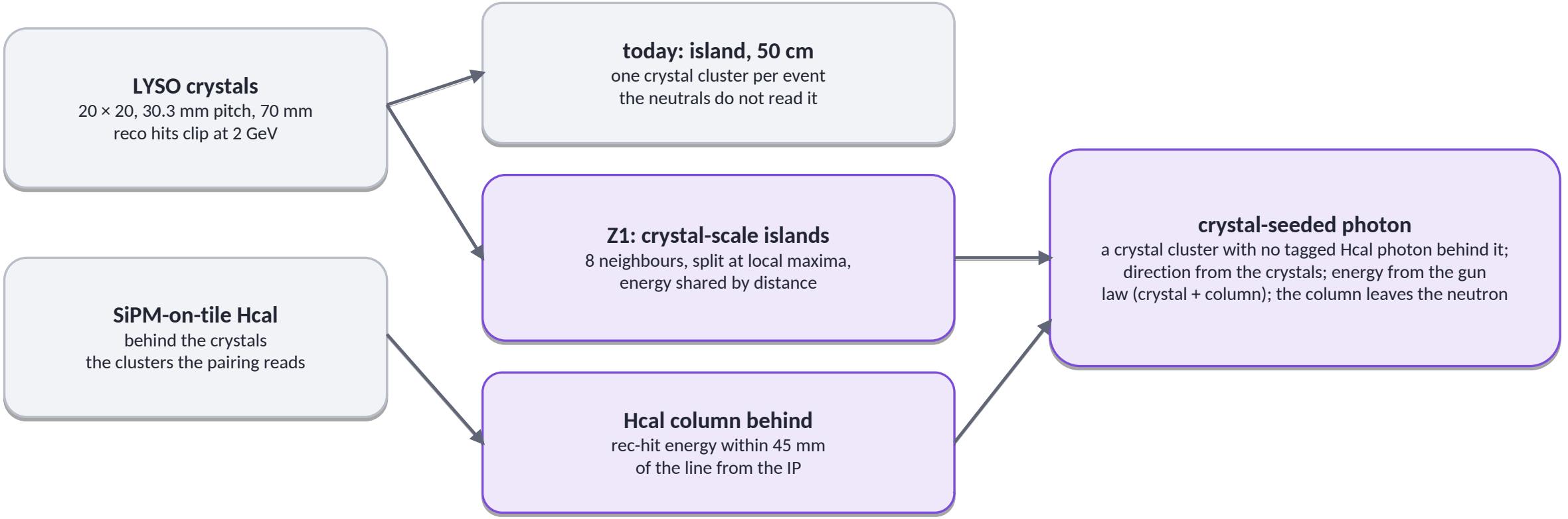
- score: N_found pooled over D6s 9x275 + 9x130
- selector 463: paired z ≥ 3, purity z_P ≥ −2
- held out: D6h, checkpoint every 5 tries
- fair play: 460 F1–F6, 465 patch scan
- new: 468, F3 for crystal clusters

−t resolution at 9x275 by true −t (GeV², σ68)

- 0.0–0.2: 0.05, 0.2–0.5: 0.08, 0.5–1.0: 0.11, 1.0–3.0: 0.18
- beam spread 0.045 GeV², detector 0.110 GeV² (476)

The all-ZDC regime holds 28 % of expected at 9x275; a ZDC photon without a photon neutral is its largest loss.

LYSO crystal section — today's clustering and a crystal-seeded photon



In the pairing

- new hypotheses beside Y9's: crystal + tagged Hcal photon, crystal + crystal, crystal + B0 photon; the lowest score wins as before

Step 0 scope study — crystal candidates for the photons Y9 misses (9x275)

ZDC photons of a $\Lambda \to n\pi^0$ (search half)	all in the ZDC	one γ in B0
without a matched photon neutral in Y9	1422	265
… with a pure crystal cluster ≥ 0.5 GeV	637	167
Λ missed with both photons available (oracle)	708	270

Emulated gain, best variant (veto split crys_pd -10)

- all-ZDC +116, one γ in B0 +7, both γ in B0 -3: net +120 found Λ
- emulator check: Y9's own decision gives 750 / 395 against the row's 749 / 395
- stop rule: below +50 pooled the phase stops here

Z1 — crystal-seeded photons in the ZDC Λ pairing

Change

- crystal section clustered at the crystal scale: 8 neighbours, split at local maxima ≥ 0.1 GeV
- a crystal cluster ≥ 0.5 GeV with no tagged Hcal photon behind it is a photon candidate
- its energy 1.962·E_crys + 47.47·E_column − 0.881 (gun fit, D2 + D2c); its column leaves the neutron
- new hypotheses: crystal + Hcal photon, crystal + crystal, crystal + B0 photon

Prediction (written before launch)

- 477 emulation: N_found 1267 → ~1385, 219 → ~234 (+133 pooled); purity 26.7 → ~28.4 %

search half	best	this try
9x275 N_found	1267	1271
9x275 purity	26.4 %	26.2 %
9x130 N_found	219	227
9x130 purity	28.7 %	29.0 %

paired: b 110, c 122, z +0.79; purity z_P -0.17

INCONCLUSIVE

fair play PASS

−t σ68 at 9x275: 0.124 GeV² (476)

Z1b — crystal photons with the Hcal column from the rec hits

Change

- Z1 summed the Hcal column over the clusters' HEXPLIT subcells: 0.45 of the rec-hit column
- its crystal photons carried a median 0.56 of their energy (Z1: +12, INCONCLUSIVE)
- Z1b sums the column over HcalFarForwardZDCRecHits (6th input), the hits the law was fitted on

Prediction (written before launch)

- as Z1's emulation: N_found 1267 → ~1385, 219 → ~234 (+133 pooled); purity 26.7 → ~28.4 %

search half	best	this try
9x275 N_found	1267	1314
9x275 purity	26.4 %	27.1 %
9x130 N_found	219	232
9x130 purity	28.7 %	29.8 %

paired: b 109, c 169, z +3.60; purity z_P +0.91

ACCEPTED | **fair play PASS**

−t σ68 at 9x275: 0.126 GeV² (476)

Z2 — crystal photon parameters: threshold, column width, neutron share

Change

- crystal candidates from 0.3 GeV (0.5 GeV)
- Hcal column 60 mm wide (45 mm), the law refitted on the guns at that width
- the column leaves the neutron on the Hcal clusters' scale (× 1/0.0273): no double counting

Prediction (written before launch)

- 477: +31 emulated over Z1b; at Z1b's 45 % fidelity ~+14: 1314 → ~1328, 232 → ~232; purity flat

search half	best	this try
9x275 N_found	1314	1388
9x275 purity	27.1 %	27.8 %
9x130 N_found	232	230
9x130 purity	29.8 %	28.7 %

paired: b 48, c 120, z +5.55; purity z_P +0.55

ACCEPTED **fair play PASS**

−t σ68 at 9x275: 0.122 GeV² (476)

Z4 — the ZDC pairing's Λ-mass width 0.04 → 0.03 GeV

Change

- with crystal candidates the chooser weighs more hypotheses per event
- 679 missed Λ at 9x275 hold pure candidates for both photons
- a narrower Λ-mass term ranks the true triplet first more often

Prediction (written before launch)

- 477 --full: +27 found, +308 in the window: 1388 → ~1409, 230 → ~236; purity 28.0 → ~27 %

search half	best	this try
9x275 N_found	1388	1409
9x275 purity	27.8 %	26.9 %
9x130 N_found	230	237
9x130 purity	28.7 %	27.9 %

paired: b 24, c 52, z +3.21; purity z_P -1.11

ACCEPTED

fair play PASS

−t σ68 at 9x275: 0.122 GeV² (476)

Z5 — the crystal law's energy as a competing hypothesis for Hcal photons

Change

- a tagged Hcal photon with a crystal cluster in front also competes with the crystal law's energy
- crystal law σ 10–11 % on the guns at 5–50 GeV; the Hcal law 24–34 %
- the chooser keeps whichever energy fits; the winner links the crystal cluster too
- Z3, the outright replacement, lost 107 Λ at 9x275 in emulation

Prediction (written before launch)

- 477 --alt: +160 over Z4 at a nearly flat window: 1409 → ~1556, 237 → ~250; purity up

search half	best	this try
9x275 N_found	1409	1574
9x275 purity	26.9 %	29.5 %
9x130 N_found	237	252
9x130 purity	27.9 %	29.7 %

paired: b 30, c 210, z +11.62; purity z_P +3.05

ACCEPTED | **fair play PASS**

−t σ68 at 9x275: 0.147 GeV² (476)

Z6 — the crystal-law photon's Hcal column leaves the neutron

Change

- Z5 measures the Λ energy high (reco / true 1.110 vs Z4's 1.073) and −t worse at 9x275
- a crystal-law Hcal photon's column beyond its own cluster stayed in the neutron: counted twice
- Z6 removes it from the neutron, as Z2 does for crystal photons

Prediction (written before launch)

- 477: Λ energy 1.122 → 1.058 for −14 found at 9x275: no gain for 463; −t below Z4's 0.122 GeV²

search half	best	this try
9x275 N_found	1574	1547
9x275 purity	29.5 %	29.0 %
9x130 N_found	252	250
9x130 purity	29.7 %	29.1 %

paired: b 68, c 39, z -2.80; purity z_P -0.58

REJECTED

fair play PASS

−t σ68 at 9x275: 0.121 GeV² (476)

Z7 — tagged photons near the neutron core pay an isolation term

Change

- The photon tag lets neutron shower fragments through; the chooser then picks one as the second photon
- Shape and depth do not tell them apart; distance to the most energetic cluster does (fragments 35 mm, photons 88 mm)
- Each tagged Hcal photon adds 2 · exp(−d / 42.5 mm) to its hypothesis's score; 42.5 mm: neutron-gun fragments

Prediction (written before launch)

- 477: +88 found over Z6 (9x275 +83, 9x130 +5), N_window −42: purity up; −t unchanged

search half	best	this try
9x275 N_found	1547	1628
9x275 purity	29.0 %	30.5 %
9x130 N_found	250	255
9x130 purity	29.1 %	29.6 %

paired: b 19, c 105, z +7.72; purity z_P +1.62

ACCEPTED **fair play PASS**

−t σ68 at 9x275: 0.120 GeV² (476)

Z8 — tagged Hcal photons may take the axis law's energy

Change

- Right pairs out of the mass window: the photons read low (Hcal law 0.69), the neutron right (1.02)
- A photon's Hcal shower splits into several clusters; the law on the leading one scatters ±30 % on guns
- Axis law: crystal + Hcal rec hits within 45 mm of the photon's line, fitted on guns (±10 %), as a competing energy

Prediction (written before launch)

- 477: +66 found over Z7 (9x275 +59, 9x130 +7), purity ~31 %; −t at or below 0.120 GeV²

search half	best	this try
9x275 N_found	1628	1688
9x275 purity	30.5 %	31.0 %
9x130 N_found	255	258
9x130 purity	29.6 %	30.0 %

paired: b 52, c 115, z +4.88; purity z_P +0.64

ACCEPTED

fair play PASS

−t σ68 at 9x275: 0.114 GeV² (476)

Λ count against −t resolution — Y9, Z5 and Z6 on the search halves

row	N_found (9x275 + 9x130)	purity	−t σ68 9x275	−t σ68 9x130	−t within ±0.1 GeV², 9x275
Y9 (start)	1486	26.7 %	0.124 GeV²	0.133 GeV²	56 %
Z5 (accepted)	1826	29.5 %	0.147 GeV²	0.128 GeV²	51 %
Z6 (the base, D-user-44)	1797	29.1 %	0.121 GeV²	0.120 GeV²	60 %

Why Z5 measures −t worse and Z6 does not

- Z5 gives tagged Hcal photons the crystal law's energy when it fits better; their Hcal column beyond the cluster stays in the neutron
- the Λ energy reads high: reco / true median 1.110 (Z4 1.073, same Λ); Z6 removes that column from the neutron
- Z6 loses 29 found Λ against Z5 (z −2.80) and measures −t better than the start row

Q22: the user chose Z6, the better −t, as the base for Z7 and Z8 (D-user-44).

Checkpoint 1 — Z5 on the held-out MSF halves and the D4 gun

N_found (yield)	Y9, search	Z5, search	Y9, held out	Z5, held out	reachable
9x275	1267 (17.70 %)	1574 (21.98 %)	1206 (16.84 %)	1520 (21.23 %)	43.6 %
9x130	219 (3.06 %)	252 (3.52 %)	205 (2.86 %)	235 (3.28 %)	8.0 %
purity, held out (pooled)			25.8 %	28.8 %	

held out: REPRODUCES

b 83, c 427, z +15.23; purity z_P +3.59

D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on; 410 credits Hcal clusters only)

- Y9: 7.44 % mean, 14.12 % at 275 GeV; Z5: 7.71 % mean, 14.56 % at 275 GeV

The held-out halves never chose anything; they confirm or revert the gain of the search halves.

Checkpoint 2 — Z8 on the held-out MSF halves and the D4 gun

N_found (yield)	Y9, search	Z8, search	Y9, held out	Z8, held out	reachable
9x275	1267 (17.70 %)	1688 (23.58 %)	1206 (16.84 %)	1621 (22.64 %)	43.6 %
9x130	219 (3.06 %)	258 (3.60 %)	205 (2.86 %)	244 (3.41 %)	8.0 %
purity, held out (pooled)			25.8 %	30.1 %	

held out: REPRODUCES

b 90, c 544, z +18.03; purity z_P +5.17

D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on; 410 credits Hcal clusters only)

- Y9: 7.44 % mean, 14.12 % at 275 GeV; Z8: 8.42 % mean, 15.99 % at 275 GeV

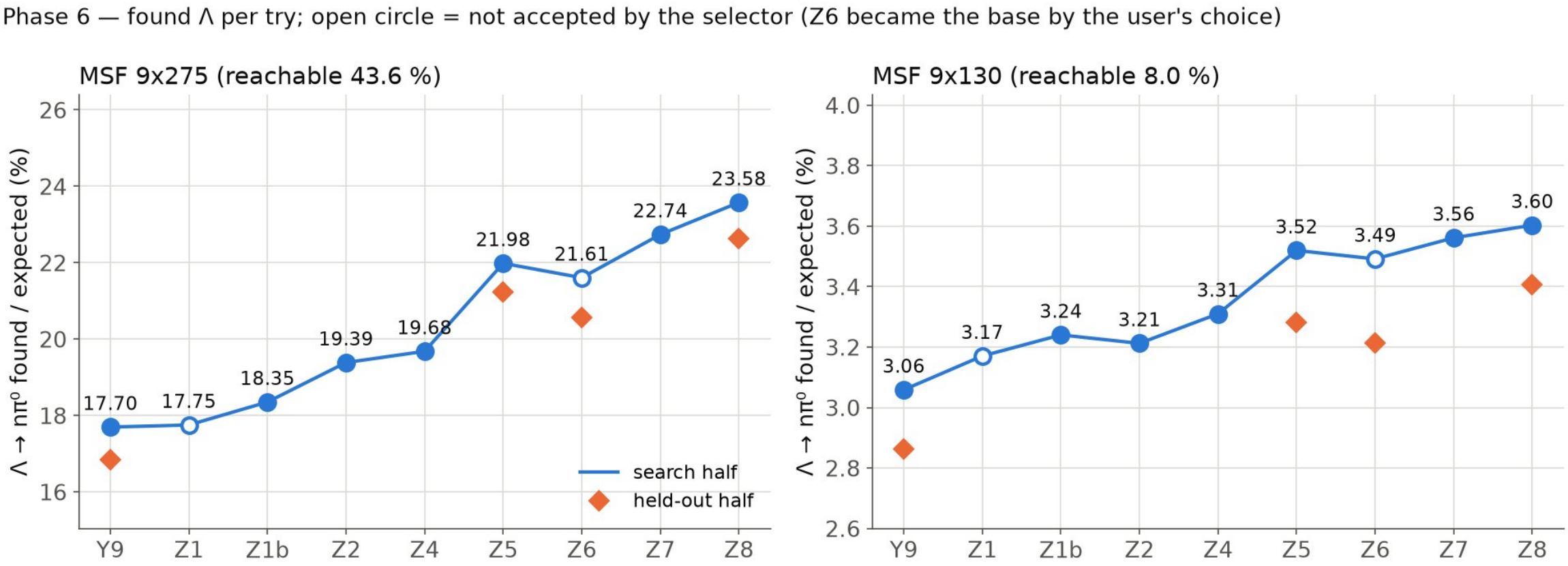
The held-out halves never chose anything; they confirm or revert the gain of the search halves.

Predictions against results — the eight Phase 6 tries

try	predicted gain (477)	measured gain (463)	verdict	reading
Z1	+133	+12	INCONCLUSIVE	the C++ column summed HEXPLIT subcells: 0.45 of the fitted column
Z1b	+133	+60	ACCEPTED	45 % of the emulated gain
Z2	+14	+72	ACCEPTED	the emulation said +31; the real chain gained more
Z4	+27	+28	ACCEPTED	held
Z5	+160	+180	ACCEPTED	held within 13 %
Z6	−14	−29	REJECTED	predicted at 9x275 only; −t repaired as predicted
Z7	+88	+86	ACCEPTED	held within 2
Z8	+66	+63	ACCEPTED	held within 3

The emulator ranks ideas; it predicts counts once its decision code mirrors the C++ (Z4 on).

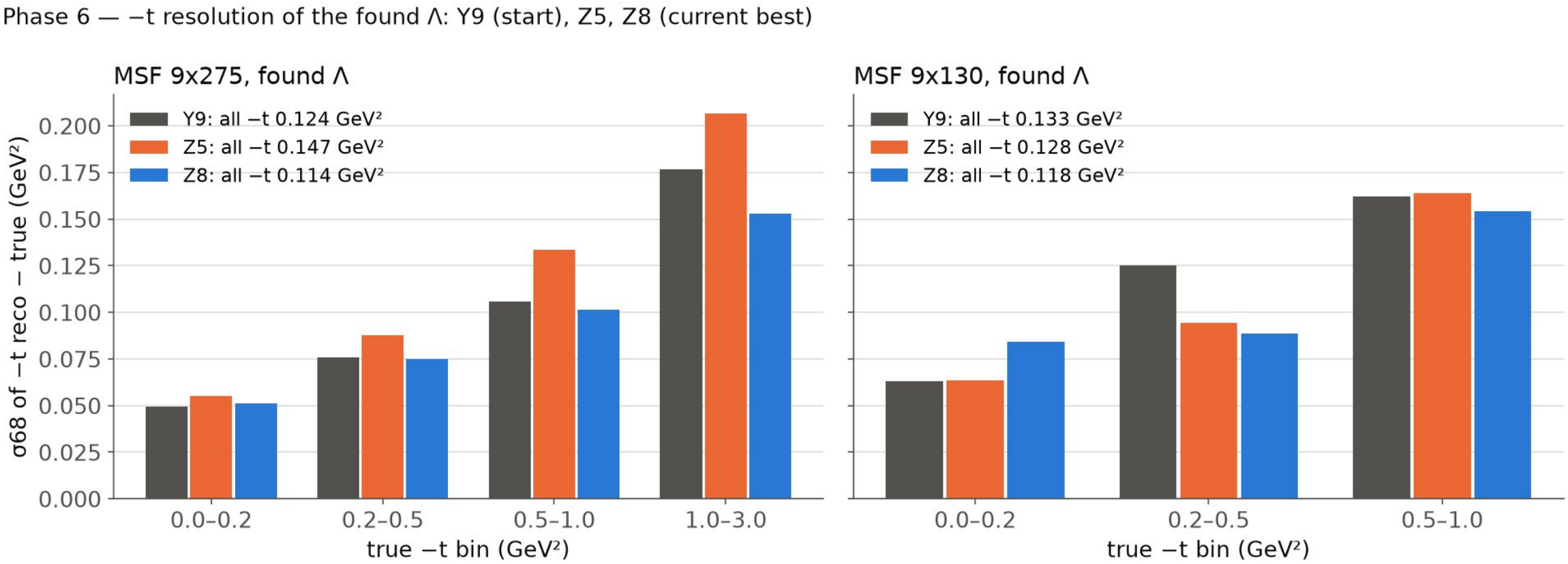
Phase 6 found Λ per try — MSF search and held-out halves



Λ → nπ⁰ found per expected (210). Filled: accepted by the selector (463); open: not accepted. Diamonds: the held-out halves D6h. Z6 is the base by the user's choice (D-user-44).

Crystal-seeded photons, the isolation term and the axis law took 9x275 from 17.7 % to 23.6 %; the held-out halves agree.

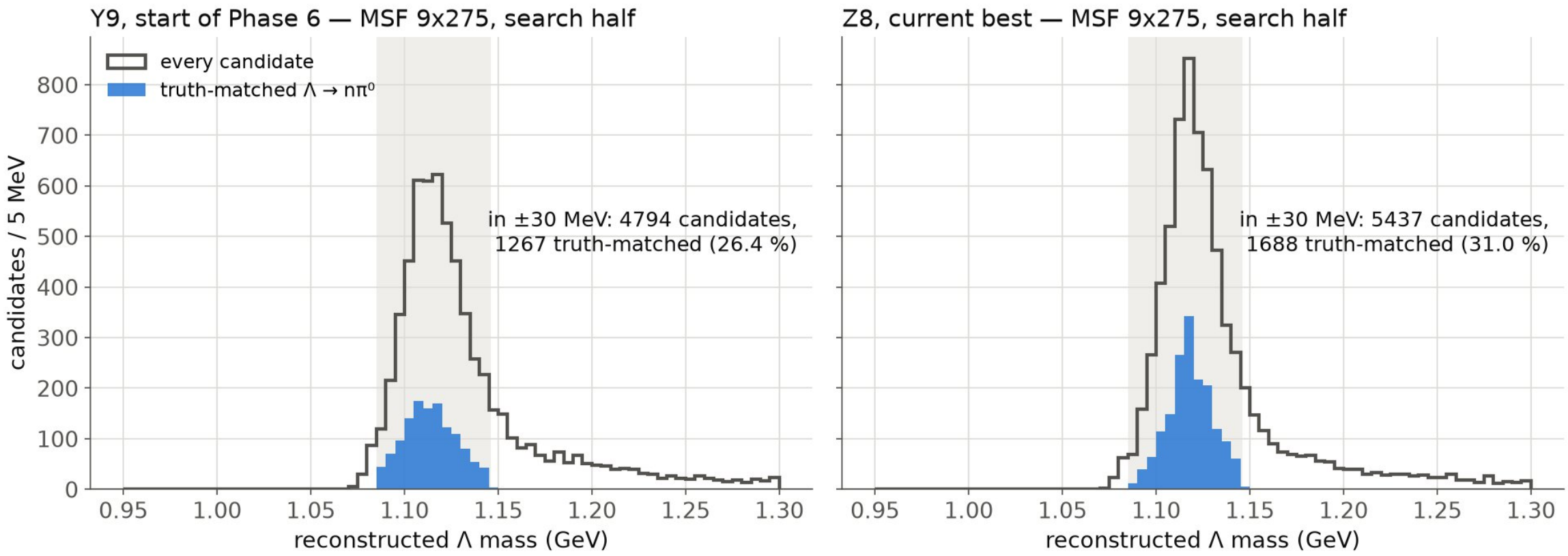
−t resolution of the found Λ per −t bin — Y9, Z5, Z8



σ68 of −t reco − true (476) on each row's found Λ, by true −t; bins with fewer than 30 Λ left out.

Z8, the current best, finds the most Λ with the best −t at 9x275; Z5 had paid for its count in −t.

Λ mass on MSF 9x275 — Y9 and Z8



Every ReconstructedLambdas mass on the 9x275 search half and the truth-matched Λ → nπ⁰ (210); band: ±30 MeV.

More truth-matched Λ in a narrower peak; the purity in the window rises from 26 % to 31 %.

Remaining losses after Z8 — what the chooser picks instead (9x275)

regime	the chooser's pick for a missed Λ whose photons both have a candidate	Λ
all in the ZDC	one photon right, the other a neutron fragment	110
	the right pair, mass out of the window (109 high)	113
	one photon right, the other a mixed Hcal cluster	93
	one photon right, the other a crystal cluster not a photon's	64
γ B0 + γ ZDC	B0 photon right, ZDC photon wrong	91
	the right pair, mass out of the window	55
	a B0 pair won	27

Tried in emulation, not built (gain under +30)

- crystal energy in front as a second term +15; isolation weight 1–4 flat; Z5's crystal-law option dropped −4
- EEMC-P photons (146 Λ in the reachable yield): soft, no pure EEMC-P photon neutral — outside the ZDC chain

No single cause above ~110 Λ remains among Λ whose photons both have a clean candidate.

Gap to the reachable yield after Z8 — where reachable Λ are lost

per expected $\Lambda \to n\pi^0$ (7160 per half)	9x275	9x130
reachable: n in the ZDC, γγ in ZDC / B0 / EEMC-P	3124 (43.6 %)	580 (8.1 %)
a photon has no clean cluster	704 (9.8 %)	200 (2.8 %)
clean clusters, another photon picked	~540 (7.5 %)	~60 (0.8 %)
the right three, mass out of the window	~200 (2.8 %)	~62 (0.9 %)
found by Z8	1686 (23.5 %)	258 (3.6 %)

No clean cluster, by cause (9x275 / 9x130)

- inside the neutron's Hcal cluster: 278 / 12
- no cluster carries it (edge clippers): 118 / 18
- both photons in one Hcal cluster: 89 / 1
- a clean cluster the photon tag rejects: 10 / 1
- B0: below 1 GeV 63 / 42; two photons in one 48 / 36; with the neutron 44 / 11
- EEMC-P photon, no pure EEMC-P neutral: 54 / 79

The largest single loss

- late decays put the photon a median 24 mm from the neutron at the Hcal face; one Hcal cluster holds both
- all-ZDC Λ decaying 30–40 m from the IP are found at 6 %, before 20 m at 61–63 %
- 53 of the 278 have a clean crystal cluster that the veto (39) or the 0.3 GeV floor (14) drops

Z8 finds 54 % of the reachable Λ at 9x275; no single remaining cause is above 6 % of the reachable yield.

Phase 5 and Phase 6 side by side

	Phase 5 — $\Lambda \to n\pi^0$ on MSF events	Phase 6 — the crystal-seeded photon finder
start → final row	X9 → Y9	Y9 → Z8
tries	9: 7 accepted, 1 inconclusive, 1 rejected	8: 6 accepted, 1 inconclusive, 1 rejected (made the base)
MSF 9x275 found / expected, search	13.2 % → 17.7 %	17.7 % → 23.6 %
MSF 9x130 found / expected, search	0.92 % → 3.06 %	3.06 % → 3.60 %
found, held-out halves	960 → 1411	1411 → 1865
purity in the window, search (pooled)	19.8 % → 26.7 %	26.7 % → 30.9 %
−t σ68 at 9x275	Y9 0.124 GeV² (X9 not measured)	0.124 → 0.114 GeV²
D4 S4a_tm per thrown, mean	7.44 % → 7.44 %	7.44 % → 8.42 %
what changed	vertex lever arm, pairing with B0 photons, B0 splitting and law	crystal-seeded photons, crystal and axis energy laws, isolation term

Phase 5 won in the B0 regimes, Phase 6 in the all-ZDC regime; both gains hold on the held-out halves.

EICrecon changes since the pin — the stack Phase 7 takes upstream

pin e5e2a60c
EICrecon at the program's pin

↓

PR 3064 + PR 3063 (open upstream)
ZDC clustering and calibration; Λ mass windows removed

↓

Phase 4: U1–U4
stable vertex, Λ-hypothesis pairing, core split, ZDC configuration

↓

Phase 5: Y2, Y5, Y9 + B0 parameters
lever-arm vertex, pairing with B0 photons, B0 splitting and law

↓

Phase 6: Z1b–Z8 + crystal clustering
crystal-seeded photons, energy laws, isolation term

Moves collections outside the ZDC

- B0ECalClusters: island splitting at local maxima (Phase 5)
- EcalFarForwardZDCClusters at the crystal scale: zdc_lyso reads them
- three new inputs of the neutrals factory in reco.cc
- ImagingTopoCluster is shared with other calorimeters: core split off by default, never measured

Not yet code

- Z8's 15 run-time parameters are not code defaults
- clang-format, units, Catch2 tests, defaults off in the Config structs

Whole diff from a15b0db6: space/notes/patches/phase6_working_tree.patch (1054 added lines, 7 files). Uncommitted; nothing pushed.

Phase 7 splits this stack into pull requests and measures what Phase 4 only argued.

Phase 7 (prepared) — the changes as EICrecon pull requests, one unit at a time

split
units in order rebuild the Phase 6 tree

→

build
clang-format, units, tests, defaults off

→

switch off
every collection identical to the base

→

other calorimetry
D5 DIS + guns: 130 identity, capybara 0

→

benchmarks
declared collections: zdc_*, zdc_lyso, B0

→

PR draft
evidence + text; the user opens it

Units proposed (Step 0 confirms the split)

- photon depth limit measured from the ZDC face (a bug fix)
- ImagingTopoCluster core split, off by default
- LambdaReconstruction vertex with lever arms
- ZDC Λ pairing; crystal-seeded photons (both off by default)
- LYSO clustering at the crystal scale; B0 ECal splitting and law
- the ZDC configuration that switches it on: reproduces Z8

Rules

- each unit declares the collections it may move; every other one stays byte-identical
- orange steps: the user's 'other calorimetry not affected'
- the cleaned-up full stack reproduces Z8 Λ by Λ
- the user opens every pull request; nothing is pushed
- start: the loop v7 line in CLAUDE.md

Open before the start: the base (EICrecon main with the open PRs 3064 and 3063, or the pin) and two benchmark samples.